

Tech & Teach

Digital Chip Design Program

RTL Design

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Sequential Datapath Design Part 1 - Blocking & Non-

Blocking

Version 1.0

Task #7.1

1.1

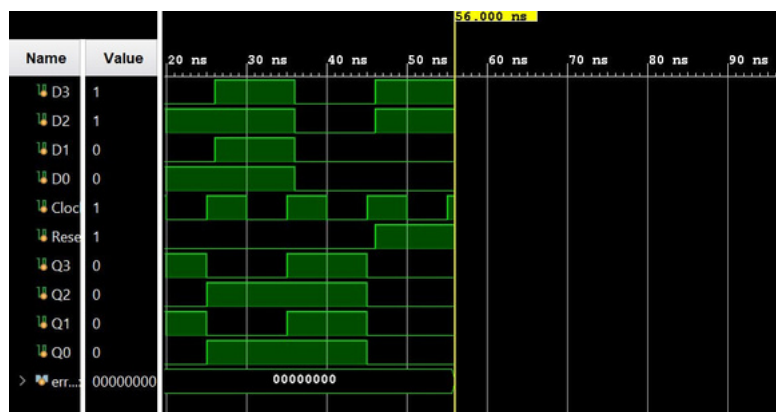
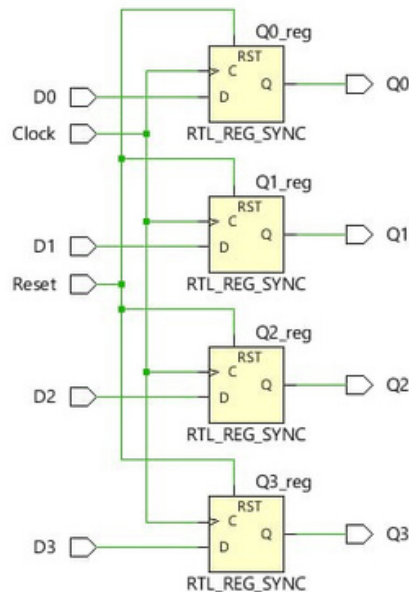
```

module reg4bit (
    input logic D3, D2, D1, D0,
    input logic Clock,
    input logic Reset,
    output logic Q3, Q2, Q1, Q0
);

always_ff @(posedge Clock) begin
    if (Reset) begin
        Q3 <= 1'b0;
        Q2 <= 1'b0;
        Q1 <= 1'b0;
        Q0 <= 1'b0;
    end
    else begin
        Q3 <= D3;
        Q2 <= D2;
        Q1 <= D1;
        Q0 <= D0;
    end
end

endmodule

```



```
`timescale 1ns/1ps
```

```
module reg4bit_tb;
```

```

    logic D3, D2, D1, D0;
    logic Clock, Reset;
    logic Q3, Q2, Q1, Q0;
    integer errors;

```

```

    reg4bit DUT (
        .D3(D3), .D2(D2), .D1(D1), .D0(D0),
        .Clock(Clock),
        .Reset(Reset),
        .Q3(Q3), .Q2(Q2), .Q1(Q1), .Q0(Q0)
    );

```

```

    initial Clock = 0;
    always #5 Clock = ~Clock;

```

```

    task check(input logic ed3, ed2, ed1, ed0, input
    string label);
        if ({Q3,Q2,Q1,Q0} != {ed3,ed2,ed1,ed0}) begin
            errors = errors + 1;
            $display("[FAIL] %s : Expected=%b%b%b%b
            Got=%b%b%b%b",
                label, ed3, ed2, ed1, ed0, Q3, Q2, Q1,
                Q0);
        end else begin
            $display("[PASS] %s : Q=%b%b%b%b", label,
                Q3, Q2, Q1, Q0);
        end
    endtask

```

```

    initial begin
        errors = 0;

```

```

        Reset = 1;
        D3 = 1; D2 = 1; D1 = 1; D0 = 1;
        @(posedge Clock); #1;
        check(0,0,0,0, "After Reset");

```

```

        Reset = 0;
        D3 = 1; D2 = 0; D1 = 1; D0 = 0;
        @(posedge Clock); #1;
        check(1,0,1,0, "Load 1010");

```

```

        D3 = 0; D2 = 1; D1 = 0; D0 = 1;
        @(posedge Clock); #1;
        check(0,1,0,1, "Load 0101");

```

```

        D3 = 1; D2 = 1; D1 = 1; D0 = 1;
        @(posedge Clock); #1;
        check(1,1,1,1, "Load 1111");

```

```

        D3 = 0; D2 = 0; D1 = 0; D0 = 0;
        @(posedge Clock); #1;
        check(0,0,0,0, "Load 0000");

```

```

        D3 = 1; D2 = 1; D1 = 0; D0 = 0;
        Reset = 1;
        @(posedge Clock); #1;
        check(0,0,0,0, "Reset during operation");

```

```

    $display("=====
    =");

```

```

        if (errors == 0)
            $display("ALL TESTS PASSED SUCCESSFULLY!");
        else
            $display("%0d TEST(S) FAILED", errors);

```

```

    $display("=====
    =");

```

```

    $finish;
    end

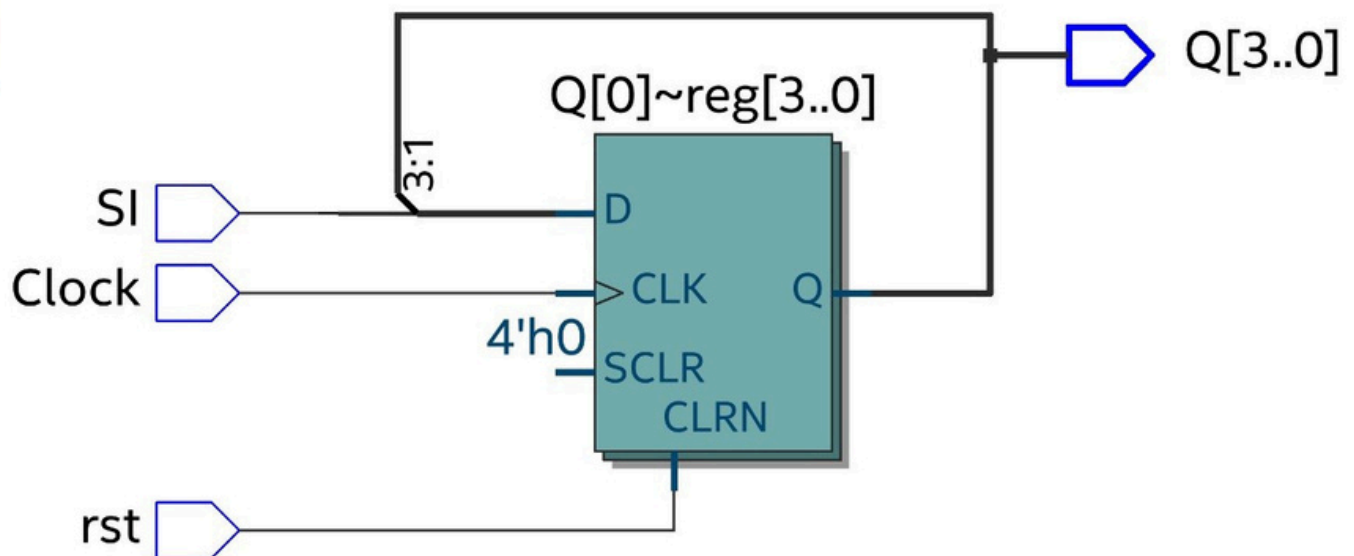
```

```
endmodule
```

Task #7.2

2.1

```
module shift_register (  
    input logic SI,  
    input logic Clock,  
    input logic rst,  
    output logic [3:0] Q  
);  
always @(posedge Clock , posedge rst) begin  
    if (rst)  
        Q <= 4'b0000;  
    else begin  
        Q[3] <= SI;  
        Q[2] <= Q[3];  
        Q[1] <= Q[2];  
        Q[0] <= Q[1];  
    end  
end  
endmodule
```



ch

2.2

```
`timescale 1ns/1ps
```

```
module shift_register_tb;
```

```
logic SI;  
logic Clock;  
logic rst;  
logic [3:0] Q;
```

```
shift_register dut (
    .SI(SI),
    .Clock(Clock),
    .rst(rst),
    .Q(Q)
);
```

always #5 Clock = ~Clock;

initial begin

Clock = 0;

```
rst = 1;
```

SI = 0;

#10;

```
rst = 0;
```

SI = 1; #10;

SI = 1; #10;

SI = 1; #10;

SI = 0; #10;

SI = 0; #10;

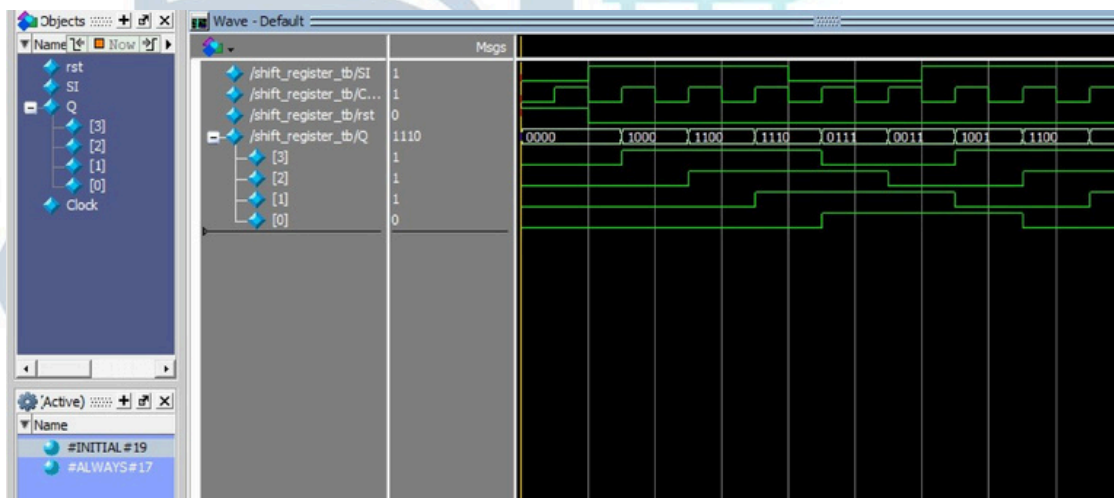
SI = 1; #10;

#20;

```
$finish;
```

end

endmodule

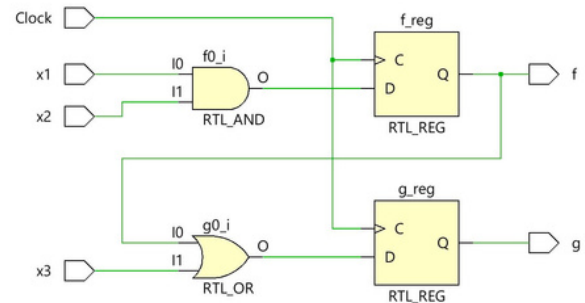
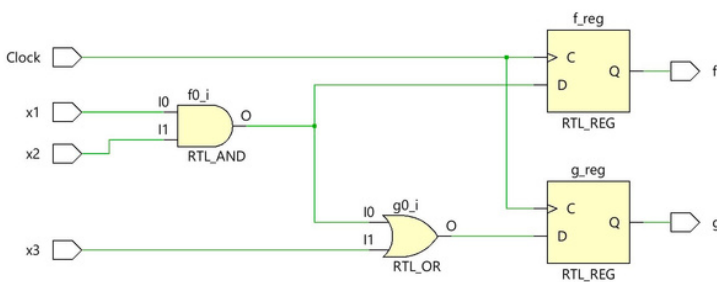


Task #7.3

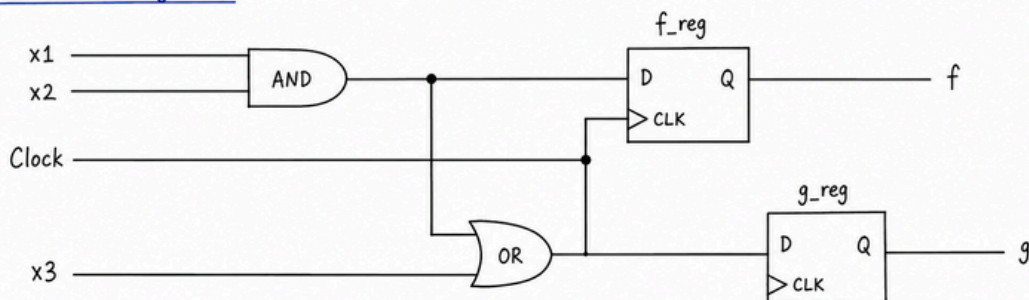
1.3

```
module example3 (
    x1, x2, x3, Clock, f, g
);
input x1, x2, x3, Clock;
output reg f, g;
always @(posedge Clock)
begin
    f = x1 & x2;
    g = f | x3;
end
endmodule
```

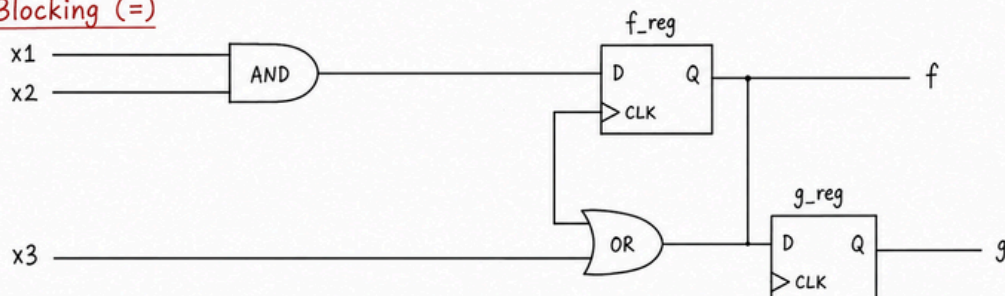
```
module example4 (
    x1, x2, x3, Clock, f, g
);
input x1, x2, x3, Clock;
output reg f, g;
always @(posedge Clock)
begin
    f <= x1 & x2;
    g <= f | x3;
end
endmodule
```



1) Non-Blocking (<=)



2) Blocking (=)



Work distribution

Section	Almas	Atheer	Mjd	Maha	Jood
Report	✓	✓	✓	✓	✓
Task6.1	✓	✓	✓	✓	✓
Task6.2	✓	✓	✓	✓	✓
Task6.3	✓	✓	✓	✓	✓

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